

Question 3 code

```
import math
X, t = 0.90, 1.0
k = -math.log(1 - X)/t
tau = X/(k*(1 - X))
print(k, tau)
```

```
2.302585092994046 3.9086503371292673
```

Question 4 code

```
import math
Xc = 0.37
Da = Xc/(1 - Xc)
X_total = 1 - (1 - Xc)*math.exp(-Da) # CSTR then PFR of same volume
def Xseries(Da_total,n): return 1 - (1 + Da_total/n)**(-n)
for n in range(1,8):
    print(n, Xseries(2*Da, n)) # total volume 2V
```

```
1 0.5401459854014599
2 0.6031000000000001
3 0.6288767463083064
4 0.6429477593021715
5 0.6518192452885667
6 0.6579259022420096
7 0.6623867186418249
```

Question 6 code

```
import math
R = 8.314 # J/kmol K = Pa m3 / kmol K
T = 523
P = 400000 # Pa
k = 2.5 # m3/kmol s
MW = 56
feed_mass = 10 * 1000 / 3600 # kg/s
F_fresh = feed_mass / MW # kmol/s
X_p = 0.8
F_A0 = F_fresh / X_p
C_A0 = P / (R * T) # kmol/m3
v0 = F_A0 / C_A0
epsilon = -0.5
def I(X): return 0.25 / (1 - X) - 0.5 * math.log(1 - X) - 0.25 + 0.25 * X
V = v0 / (k * C_A0) * I(X_p)
print(V)
```

```
5.875386066291072e-06
```

Double-click (or enter) to edit

